Fairness report

Does Kept treat protected groups differently? (compliance-only inputs)

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TL;DR

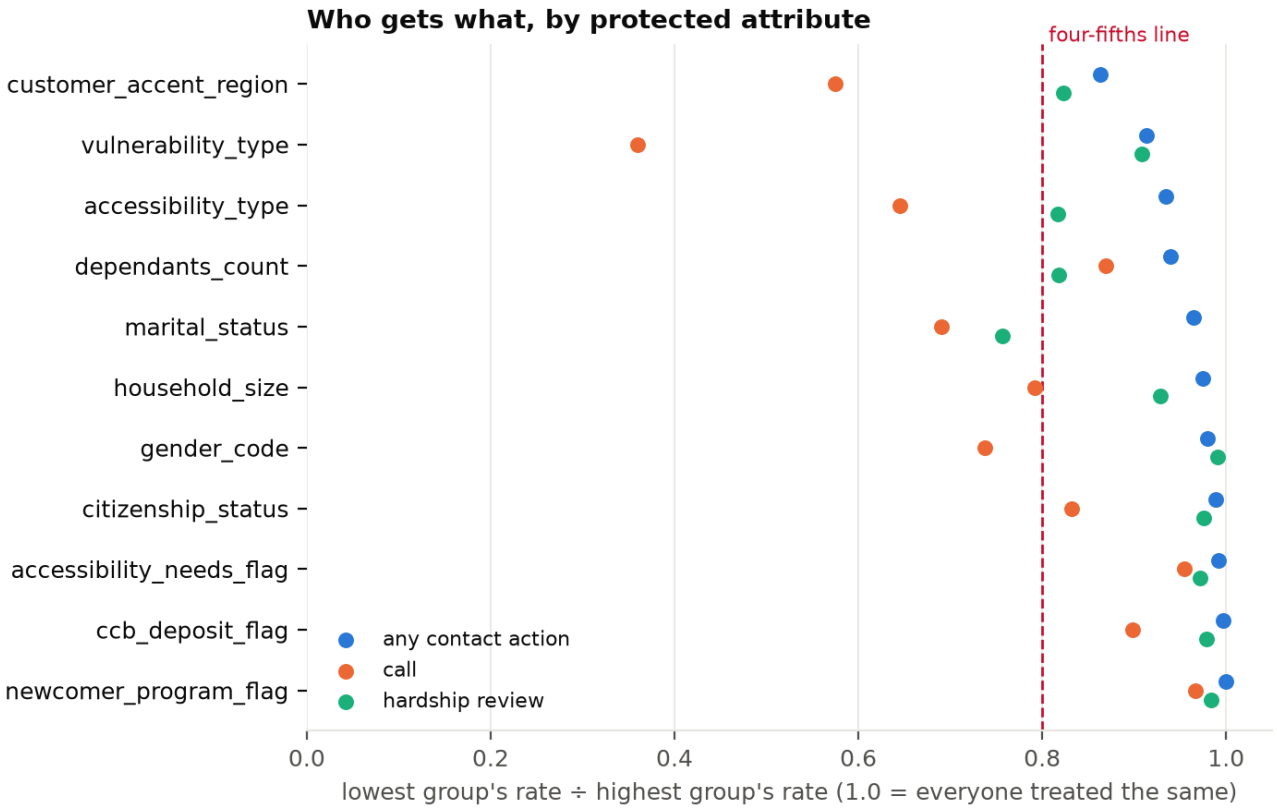
No protected group gets materially different treatment. Across 96,000 scored cases, the share of customers who get *any* contact action stays within a ratio of **0.86–1.00** between the lowest and highest group of every protected attribute (the four-fifths rule flags below 0.80).

The flags you'll see sit on **calls**, a rare action (~3% of cases), where tiny rates make ratios jumpy, and on **hardship review**, which is *supposed* to land more on vulnerable customers. That's the protection doing its job.

How this is possible: protected attributes never go into a decision. They live in `gold.c360_restricted`, which only compliance can read; a CI test fails the build if any model feature's lineage touches one (or a proxy like postal code, age or language); and a curated test suite checks the trained model's actual inputs. We use them for exactly one thing: this report.

The overview

Each dot is the lowest group's rate divided by the highest group's rate for one attribute and one outcome. 1.0 means every group gets that outcome equally often.

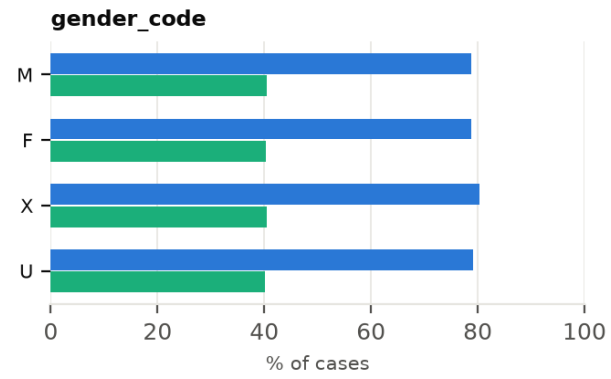


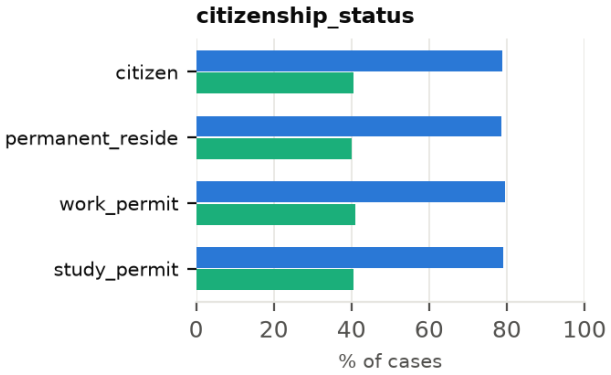
Attribute	Any contact	Call	Hardship review	Cases with it blank
customer_accent_region	0.86	0.57 (flag)	0.82	94,786
vulnerability_type	0.91	0.36 (flag)	0.91	82,304
accessibility_type	0.93	0.64 (flag)	0.82	93,394
dependants_count	0.94	0.87	0.82	30,681
marital_status	0.96	0.69 (flag)	0.76 (flag)	10,685
household_size	0.98	0.79 (flag)	0.93	30,741
gender_code	0.98	0.74 (flag)	0.99	4,389
citizenship_status	0.99	0.83	0.98	4,391
accessibility_needs_flag	0.99	0.95	0.97	3,458
ccb_deposit_flag	1.00	0.90	0.98	51,136
newcomer_program_flag	1.00	0.97	0.98	3,458

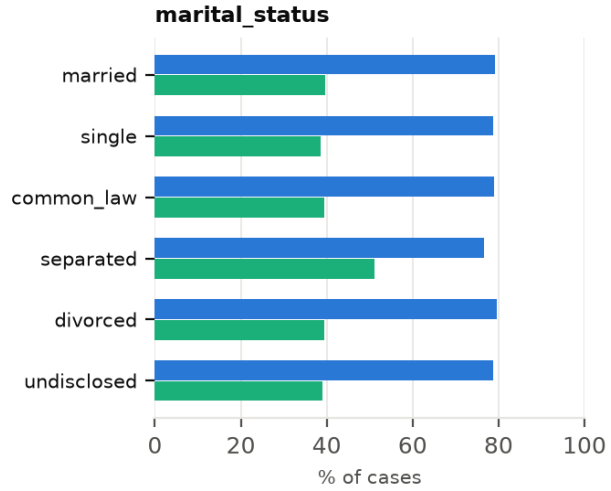
(flag) = below 0.80. A flag is a question for compliance, not a verdict.

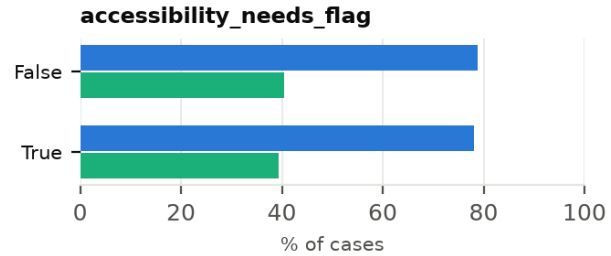
Group by group

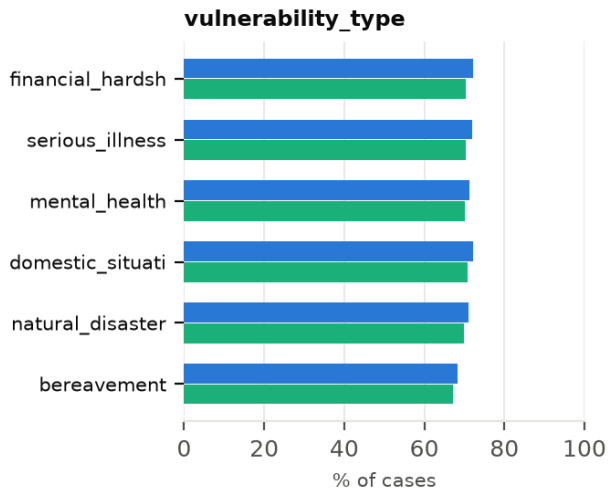
Blue = gets any contact action, aqua = gets a hardship review.

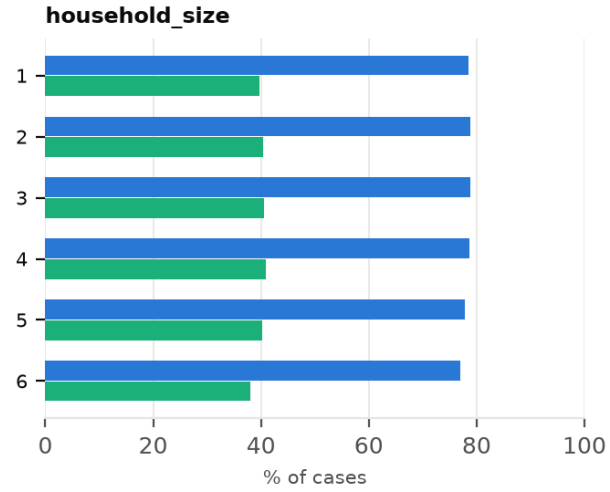

gender_code
M
F
X
U
0
20
40
60
80
100
% of cases


citizenship_status
citizen
permanent_reside
work_permit
study_permit
0
20
40
60
80
100
% of cases


marital_status
married
single
common_law
separated
divorced
undisclosed
0
20
40
60
80
100
% of cases


accessibility_needs_flag
False
True
0
20
40
60
80
100
% of cases





What we read from it

1. **Contact is even.** Gender, citizenship, household, dependants, benefits and accessibility all sit at 0.93–1.00 for "any contact action".
2. **Call flags are small-number noise.** Calls are ~3% of cards; 3.5% vs 2.9% is a 0.83 ratio but a tiny absolute gap.
3. **Vulnerable customers get more hardship reviews, on purpose.** The vulnerability *flag* (a safeguard, not a protected attribute in the data dictionary) routes cases to protective actions. That shows up as a gap by `vulnerability_type`, and it's the point.
4. **Blank attributes go with fewer contacts.** That's thin data (often holds, cease or the hold-out), not a group. Worth a data-coverage look, not a fairness fix.

Rebuild: `uv run python scripts/fairness_report_pdf.py` (with `kept serve` stopped, on a compliance machine).